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Is prayer an effective intervention for pain management? A protocol of a systematic review of randomized controlled trials

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Is prayer an effective intervention for pain management? A protocol of a systematic review of randomized controlled trials

Alexandra Ferreira-Valente[1,2], Margarida Jarego[1], Inês Queiroz-Garcia[1], Filipa Pimenta[1], Rui Miguel Costa[1], Melissa A. Day[2,3], José Pais-Ribeiro[1], & Mark P. Jensen[2]

[1] William James Center for Research, ISPA – University Institute, Lisbon, Portugal

[2] University of Washington, Department of Rehabilitation Medicine, Seattle, WA, USA

[3] University of Queensland, School of Psychology, Faculty of Health & Behavioral Sciences, St. Lucia Campus, Brisbane, QLD 4072, Australia

Corresponding Author:

Alexandra Ferreira-Valente, PhD

William James Center for Research, ISPA – University Institute, Rua Jardim do Tabaco, n.º 34, 1149-041 Lisbon, Portugal

Tel.: +351 969082988

Email: mafvalente@gmail.com

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ABSTRACT

Background: Pain is a universal experience and the most common reason for seeking health care. Inadequate pain management negatively impacts numerous aspects of patient health. Multidisciplinary treatment programs, including psychosocial interventions, are more useful for pain management than purely biomedical treatment alone. Recently, researchers showed increasing interest in understanding the role of spirituality/religiosity and spiritual/religious practices on pain experience, with engagement in religious practices, such as prayer, showing to positively impact pain experience in religious individuals. This systematic review will seek to summarize and integrate the existing findings from randomized controlled trials assessing the effects of prayer and prayer-based interventions on pain experience.

Methods: The systematic review procedures and its report will follow the PRISMA statement checklist. Electronic searches in 9 databases (Web of Science Core Collection, MEDLINE, SCIELO Citation Index, PUBMED, Cochrane Central Register of Controlled Clinical Trial, PsycINFO, Scopus, LILACS, and Open-SIGLE) will be performed to identify randomized controlled trials of prayer-based interventions. Two independent researchers will assess studies for inclusion and extract data from each paper. The risk of bias assessment will be assessed independently by two reviewers based on the CONSORT statement. Qualitative synthesis of the body of research will be conducted using a narrative summary synthesis method. Meta-analysis will be limited to studies reporting on the same primary outcome.

Discussion: Findings will be useful to (1) understand the condition of our knowledge in this field; and (2) provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain.

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Systematic review registration: This protocol was submitted for registration at the International Prospective Register for Systematic Reviews (PROSPERO) number 221733.

Keywords: Pain; Prayer; Systematic Review; Protocol

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is comprehensive systematic review focusing the effects of prayer (despite the type of prayer) on pain related-outcomes (despite the type of pain), including studies published in peer-reviewed indexed journals and on grey literature regardless of the date of publication
- PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) checklist guidelines will be followed to conduct this review, and PRISMA-P checklist was followed to report the protocol of this systematic review, to ensure quality in all aspects of study planning, execution and reporting.
- Risk of bias assessment, assessment of reporting and publication bias, and evaluation of the confidence in cumulative evidence will be performed, respectively, based on CONSORT (Consolidated Standards of Reporting Trials) statement, Egger's regression test method, and on the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.
- Main strengths of the study are its comprehensiveness, including studies and grey literature published in five languages despite the date of publication, and the use of well-established and standardized methods to perform qualitative synthesis, meta-analysis, risk of bias assessment and meta-bias assessment.
- We may not be able to conduct the anticipated meta-analysis due to methodological heterogeneity between the included studies and between the primary outcomes' measures used in the different studies.

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INTRODUCTION

Pain is a universal experience (1). Despite its unpleasantness, pain – except when associated with many chronic pain conditions – has an adaptive purpose. It is a sign that physical damage might be occurring that threatens the well-being of a given organism (2). Untreated acute pain – pain associated with a noxious stimulus that has a sudden onset and lasts for a limited period, remitting when the underlying cause resolves (3) – often produces acute neurohumoral changes (4). Pain is the most common reason for seeking health care (5), and inadequate management of acute pain negatively impacts numerous aspects of patient health. This includes reduced quality of life, impaired sleep, impaired physical function, and increased risk of developing chronic pain (6). As a result, pain remains a significant health issue, and its management is a priority worldwide (4,6–8).

Since pain is a multidimensional and subjective experience, researchers and clinicians have worked to clarify the role that biological (e.g., hormones, the extent of physical damage), psychological (e.g., beliefs, coping responses), and social factors (e.g., social support) have in influencing the experience and impact of pain (9). This research body informed the development of some pain management multidisciplinary treatment programs, including psychosocial interventions (10–12), which are more effective in managing chronic pain than treatments that focus only on biomedical factors (9,13,14).

More recently, researchers have become interested in understanding the association between pain, adjustment to pain, spirituality (and spiritual practices), and religiosity (and religious practices). Taken together, findings regarding the use of spiritual/religiosity and prayer-based interventions for pain are encouraging. Preliminary findings indicate that people who are more engaged with meaningful

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spiritual practice can better tolerate pain (15,16), i.e., the maximum intensity of a pain-producing stimulus that a person is willing to accept in a particular situation (17). Other research (18) with Muslim women who underwent a caesarean section showed that compared to a no treatment control, women randomized to an Islamic prayer condition showed significantly lower pain intensity scores, i.e., the magnitude of the experienced pain (19), at three hours and six hours after the intervention (18). However, there were no significant differences in objective stress responses, i.e., physiological responses, such as blood pressure, respiration, and heart rate (20).

Two recent systematic reviews summarized the evidence relative to the extent to which measures of religiosity/spirituality are associated with pain-related outcomes, pain-related beliefs, and pain coping responses in people with chronic pain (21), and to the benefits of active prayer to a deity as adjunctive therapy for pain (22). Results from these reviews suggested that while spirituality/religiosity are associated with better physical and psychological function in people with chronic pain (21), within the acute pain context, active prayer to a deity appears to be useful in reducing pain intensity and increasing pain tolerance in religious people – but not in non-religious people – undergoing surgery or a painful procedure (22). Although one of these systematic reviews focused specifically on the effects of prayer to a deity, this systematic review was limited to studies published in English between the years 2000 and 2019, in peer-reviewed journals indexed in one or more of only four databases (PubMed, Web of Science, PsychINFO, and Scopus). Thus, this review might have missed an important number of studies that: (a) were published in languages other than English, (b) were published before the year 2000, and (c) were published in journals indexed in other relevant databases or present in the grey literature. This systematic review was also restricted to the effects of *individual* prayer on pain, thus excluding studies focusing on

community, shared, or group prayer, which is a religious practice of many religions. Furthermore, this previous review did not provide: (a) a qualitative review of the findings based on a standardized and well-established method for qualitative synthesis of the findings; (b) did not describe the characteristics of the participants in the studies reviewed; and (c) did not conduct a quality assessment of the studies included in the review. A more comprehensive systematic review focusing the effects of both individual and community, shared or group prayer on acute and chronic pain related-outcomes, including studies published both in peer-reviewed indexed journals and on grey literature regardless of the date of publication, using a standardized method for performing qualitative data analysis and metanalysis, and assessing both included studies characteristics and their methodological quality, would be useful for a more comprehensive and complete understanding of the usefulness of prayer as a pain management intervention. Hence, the present study aims to perform such a systematic review.

Study aims

The current systematic review seeks to summarize and integrate the existing findings from randomized controlled trials assessing the effects of prayer-based interventions (including individual and share/group prayer) on pain-related outcomes (i.e., pain intensity, pain tolerance, and pain-related stress). Specifically, this systematic review seeks to examine the extent to which prayer affects pain-related outcomes in adults with acute or chronic pain while including studies published in English, German, French, Spanish, Italian or Portuguese, searched in nine databases (including one grey literature database), performing qualitative data analysis and studies' methodological quality assessment using standardized and well-established methods, and making recommendations for future research in this area.

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METHOD

Review protocol and registration

The present systematic review will be conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement for systematic reviews and meta-analysis protocols (23). Likewise, the PRISMA-P checklist was followed to report the protocol of this systematic review (24). This review will be registered in the PROSPERO registry before the review is conducted (number 221733).

Review participants

Participants will be adult individuals experiencing pain, regardless of the cause of pain, pain duration, or pain location.

Eligibility criteria

Articles will be included in this systematic review if: (1) participants are adults (i.e., 18 years old or older); (2) participants experienced pain, regardless of its duration or etiology; (3) they include quantitative measure(s) of pain intensity, pain tolerance, and/or pain-related stress; (4) they assessed prayer-based interventions; (5) they compared the intervention group to a control group (e.g., alternative treatment(s), treatment as usual, or both); (6) used a randomized controlled trial design; (7) they were published in English, German, French, Spanish, Italian or Portuguese before October 3, 2020. While including only studies written in these languages might be restrictive, the authors are only proficient in these languages and do not have the resources to include articles written in other languages in the screening or the review. Papers reporting the results of qualitative, cohort, case-control, and cross-sectional studies and papers focusing on spiritual practices or Complementary and Integrative Health interventions

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other than prayer (e.g., meditation, Reiki, and yoga), will be excluded from the systematic review.

Patient and public involvement

No patient is involved in either the design or planning phase of this study.

Search strategy

The systematic review protocol specifies and documents the search strategy, methods, and inclusion/exclusion criteria of the articles in advance. Search strategies to use to identify relevant publications will be customized to suit each database. Search terms to use in each database to search for publications regarding the previously mentioned pain-related outcomes will be (1) Religion OR Prayer AND (2) Pain AND (3) Pain intensity OR Pain tolerance OR Stress. The search terms will be used in various combinations, considering synonyms and MESH entries (*cf*. Supplementary file 1).

We will search 9 databases for the relevant publications: Web of Science Core Collection (Clarivate Analytics), MEDLINE (via Web of Science, Clarivate Analytics), SCIELO Citation Index (via Web of Science, Clarivate Analytics), PUBMED, Cochrane Central Register of Controlled Clinical Trial (via OvidSP), PsycINFO, Scopus, LILACS, and Open-SIGLE. Unpublished literature will also be searched in clinical trial registry platforms, such as ClinicalTrials.gov and ISRCTN registry. Then, to identify additional articles not identified during the database search, references list of articles included in the review and review articles found by the search will be hand searched.

Co-Primary outcomes

Studies should use at least one quantitative measure of pain intensity, pain tolerance, or pain-related stress to be included in the systematic review. The following are the commonly used measures assessing these variables.

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Among the most common measures of pain intensity used by researchers and clinicians are the Visual Analogue Scale [VAS (25)] and related versions of this scale (e.g., mechanical VAS, digital VAS). Other commonly used measures of pain intensity (26) are the 0-10 Numerical Rating Scale (NRS), the Verbal Rating Scale (VRS), and the Faces Pain Scale-Revised [FPS-R (27)]. Each of these measures is adequately valid and reliable as a measure of pain intensity, and patients in Western countries tend to prefer VRS and NRS over VAS (26). Pain tolerance is commonly operationalized as the length (in seconds) that an individual is willing to experience an aversive stimulation (28–30).

Finally, perceived stress is usually assessed with the: (a) Perceived Stress Scale [PSS (31)], a 10-item scale that asks about people's feelings and thoughts regarding the passing month, on a scale from 0 (never) to 4 (very often); (b) stress scale of Depression, Anxiety, and Stress Scale [DASS (32)], a 14-item scale where subjects are asked to use 4-point severity/frequency scales to rate the extent to which they have experienced each state over the past week; and (c) 21-item version (DASS-21), which consist of a 7-item scale similar to the previous. In contrast, objective stress is usually measured through salivary cortisol, heart rate variability, or blood pressure measurement and analysis (33–35).

Study selection, data management and data extraction

First, all references identified during the database search will be inserted into Zotero Software (36). Second, cross-references and duplicates will be deleted. Then, two independent reviewers (MJar and IG) will screen the title, abstract, and keywords of the identified studies for eligibility screening based on the eligibility criteria listed above. Full texts of the manuscripts meeting the eligibility criteria and those with an uncertainty of inclusion from the abstracts will be independently read by the same

reviewers to confirm inclusion or exclusion. Discrepancies found will be settled during a consensus meeting. In cases in which consensus is not achieved, a third reviewer will be consulted (AFV). The studies' inclusion or exclusion will be recorded on a self-designed data extraction form in Microsoft Excel 10.

Third, detailed data from the studies meeting the inclusion criteria will be extracted on a Microsoft Excel extraction sheet by two authors (MJar and IG). Data to be extracted include are: (1) authors and year of publication; (2) country of study; (3) study participant (type of pain – chronic or acute – and pain etiology); (4) sample size; (5) study participants age (mean, standard deviation); (6) percentage of female participants; (7) site of pain; (8) length and intensity (frequency of sessions) of prayer-based intervention; (9) type of prayer (10) type and characteristics of the comparator or control condition; (11) pain intensity; (12) pain tolerance; and (13) pain-related stress. Data (mean, standard deviation, or effect sizes) on the primary outcomes will be extracted. Any discrepancies in the data collected will be resolved by discussion and consensus. In the event that consensus is not achieved, a third author (AFV) will be consulted.

Risk of bias assessment

The risk of bias of the studies included in the present systematic review will be evaluated based on the CONSORT (Consolidated Standards of Reporting Trials) statement. Risk of bias will be assessed by two independent reviewers (MJar and IG) and discrepancies will be settled during a consensus meeting. If a consensus is not reached, a third reviewer (AFV) will be consulted.

Data analysis plan

Qualitative synthesis. A qualitative synthesis of the body of research will be conducted (37), describing the methodological characteristics of the studies included in this

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systematic review, their strengths and limitations, and each study's results regarding the effects of prayer in reducing pain intensity and pain-related stress and in increasing pain tolerance in adults experiencing pain, compared to treatment as usual or no treatment.

To summarize the evidence, we will implement a narrative summary synthesis method based on the framework developed by the UK Economic and Social Research Council to the conduct of narrative synthesis in systematic reviews (38), modified to fit our purposes. Following the referred guidance, the synthesis process will involve three elements: (1) developing a preliminary synthesis of findings of included studies regarding the two topics; (2) exploring relationships in the data; and (3) assessing the robustness of the synthesis.

Meta-analysis. If at least two included articles present data on a single primary outcome, then meta-analysis will be performed (39). End-point scores will be computed as expressed mean differences or standardized mean difference (SMD) for each study for primary outcomes, with associated 95% confidence intervals. Data extracted in the articles will be transformed and presented in Cohen's *d* effect size with 95% confidence interval (*CI*) when possible using the Campbell Collaboration online calculator (40). Cohen's *d* effect size thresholds to be used are: small (0.20), medium (0.50), large (0.80), and very large (1.30) (41).

Assessment of reporting biases

We will use funnel plots to evaluate the potential reporting bias and small-study effects when there are more than 10 studies in the meta-analysis. Quantification of publication bias will be done using Egger's regression test method (42).

Confidence in cumulative evidence

Included studies will be critically appraised and synthesized in terms of methodological quality along with the relevance, strength and limitations of the

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evidence presented. The similarities and differences between the studies (study characteristics, design and execution) will be described and their potential impact on study outcome will be discussed. The strength of the body of evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

DISCUSSION

The present systematic review will provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain. This allows for a more considerate and insightful choice when advising pain patients with effective coping strategies.

ETHICS AND DISSEMINATION

Ethics approval is not required for this review because it relies on secondary data. Any modification to the systematic review protocol will be submitted to review and approval of PROSPERO registry and described in the final report of the systematic review. The systematic review findings will be published in a peer-reviewed indexed journal and presented at a pain international scientific meeting.

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AUTHORS' CONTRIBUTIONS

Concept and design: AFV conceived the study idea, and MJar, MJ and MD contributed to the planning and design of the protocol.

Drafting of the manuscript: AFV and MJar wrote the first draft of the manuscript, and all the remaining authors contributed to revise the drafts and approved and contributed to the final written manuscript.

Critical revision of the manuscript for important intellectual content: All authors.

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COMPETING INTERESTS

None to declare.

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Appendix 1. Search terms, alternative search terms, and search strategy for Medline via Web of Science

	PAIN	PRAYER-BASED INTERVENTIONS	PAIN-RELATED OUTCOMES
MeSH heading	- Pain - Chronic pain - Acute pain	- Religion - Spiritual therapies	- Pain intensity - Pain tolerance - Stress
Title, abstract, and keywords	- Pain	- Religion - Pray* - Religious* - Spiritual therapies - Noetic therapies	- Pain intensity - Pain severity - Pain tolerance - Stress*

To identify articles related to prayer-based interventions

Pain

AND

Religion OR Pray* OR Religious* OR Spiritual therapies OR Noetic therapies

AND

Pain intensity OR Pain severity OR Pain tolerance OR Stress*

Limits: Adults >18 years; Humans; Randomized Control Trial

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - **Moher D, Stewart L & Shekelle P:** Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	X	☐	Pg 1, ln 1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	X	☐	Pg 3, ln 2 to 3 Pg 8, ln 8 to 9
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	X	☐	Pg 1, ln 4 to 17
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	X	☐	Pg 19, ln 1 to 9
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	X	
Support					
Sources	5a	Indicate sources of financial or other support for the review	X	☐	Pg 19, ln 11 to 16
Sponsor	5b	Provide name for the review funder and/or sponsor	X	☐	Pg 19, ln 11 to 16
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	X	☐	Pg 19, ln 11 to 16
INTRODUCTION					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Rationale	6	Describe the rationale for the review in the context of what is already known	X	☐	Pg 5, ln 1 to Pg 7, ln 17
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	X	☐	Pg 7, ln 15 to 25
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	X	☐	Pg 8, ln 13 to Pg 9, ln 2
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	X	☐	Pg 8, ln 20 to 21 Pg 9, ln 13 to 21
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	X	☐	Pg 9, ln 5 to 21 Supplementary file 1
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	X	☐	Pg 10, ln 19 to Pg 11, ln 4
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	X	☐	Pg 10, ln 19 to Pg 11, ln 4
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	X	☐	Pg 11, ln 5 to 16
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	X	☐	Pg 9, ln 22 to Pg 10, ln 18 Pg 11, ln 5 to 16
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	X	☐	Pg 9, ln 24 to Pg 10, ln 18 Pg 11, ln 5 to 16

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	X	☐	Pg 11, ln 17 to 22
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	X	☐	Pg 12, ln 11 to 18
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	X	☐	Pg 12, ln 11 to 18
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☐	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	X	☐	Pg 11, ln 23 to Pg 12, ln 10
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	X	☐	Pg 13, ln 19 to 22
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	X	☐	Pg 13, ln 23 to Pg 14, ln 5

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Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials

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Keywords:	PAIN MANAGEMENT, COMPLEMENTARY MEDICINE, STATISTICS & RESEARCH METHODS

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Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials

Alexandra Ferreira-Valente[1,2], Margarida Jarego[1], Inês Queiroz-Garcia[1], Filipa Pimenta[1], Rui Miguel Costa[1], Melissa A. Day[2,3], José Pais-Ribeiro[1], & Mark P. Jensen[2]

[1] William James Center for Research, ISPA – University Institute, Lisbon, Portugal

[2] University of Washington, Department of Rehabilitation Medicine, Seattle, WA, USA

[3] University of Queensland, School of Psychology, Faculty of Health & Behavioral Sciences, St. Lucia Campus, Brisbane, QLD 4072, Australia

Corresponding Author:

Alexandra Ferreira-Valente, PhD

William James Center for Research, ISPA – University Institute, Rua Jardim do Tabaco, n.º 34, 1149-041 Lisbon, Portugal

Tel.: +351 969082988

Email: mafvalente@gmail.com

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ABSTRACT

Background: Pain is a universal experience and the most common reason for seeking health care. Inadequate pain management negatively impacts numerous aspects of patient health. Multidisciplinary treatment programs, including psychosocial interventions, are more useful for pain management than purely biomedical treatment alone. Recently, researchers showed increasing interest in understanding the role of spirituality/religiosity and spiritual/religious practices on pain experience, with engagement in religious practices, such as prayer, showing to positively impact pain experience in religious individuals. This systematic review will seek to summarize and integrate the existing findings from randomized controlled trials assessing the effects of prayer and prayer-based interventions on pain experience.

Methods: The systematic review procedures and its report will follow the PRISMA statement checklist. Electronic searches in 9 databases (Web of Science Core Collection, MEDLINE, SCIELO Citation Index, PUBMED, Cochrane Central Register of Controlled Clinical Trial, PsycINFO, Scopus, LILACS, and Open-SIGLE) will be performed to identify randomized controlled trials of prayer-based interventions. Two independent researchers will assess studies for inclusion and extract data from each paper. The risk of bias assessment will be assessed independently by two reviewers based on the CONSORT statement. Qualitative synthesis of the body of research will be conducted using a narrative summary synthesis method. Meta-analysis will be limited to studies reporting on the same primary outcome.

Discussion: Findings will be useful to (1) understand the condition of our knowledge in this field; and (2) provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain.

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Systematic review registration: This protocol was submitted for registration at the International Prospective Register for Systematic Reviews (PROSPERO) with the reference CRD42020221733.

Keywords: Pain; Prayer; Systematic Review; Protocol

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is comprehensive systematic review focusing the effects of prayer (despite the type of prayer) on pain related-outcomes (despite the type of pain), including studies published in peer-reviewed indexed journals and on grey literature regardless of the date of publication
- PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) checklist guidelines will be followed to conduct this review, and PRISMA-P checklist was followed to report the protocol of this systematic review, to ensure quality in all aspects of study planning, execution and reporting.
- Risk of bias assessment, assessment of reporting and publication bias, and evaluation of the confidence in cumulative evidence will be performed, respectively, based on CONSORT (Consolidated Standards of Reporting Trials) statement, Egger's regression test method, and on the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.
- Main strengths of the study are its comprehensiveness, including studies and grey literature published in five languages despite the date of publication, and the use of well-established and standardized methods to perform qualitative synthesis, meta-analysis, risk of bias assessment and meta-bias assessment.
- We may not be able to conduct the anticipated meta-analysis due to methodological heterogeneity between the included studies and between the primary outcomes' measures used in the different studies.

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INTRODUCTION

Pain – defined by the International Association for the Study of Pain as an "unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage" –[1] is a universal experience. [2] Despite its unpleasantness, pain – except when associated with many chronic pain conditions – usually has an adaptive purpose. It is a sign that physical damage might be occurring that threatens the organism's well-being.[3] Pain is the most common reason for seeking health care,[4] and inadequate management of acute pain negatively impacts numerous aspects of patient health. This includes reduced quality of life, impaired sleep, impaired physical function, and increased risk of developing chronic pain.[5] In short, pain remains a significant health issue, and its management a priority worldwide.[5–8]

Because pain is a multidimensional and subjective experience, researchers and clinicians have worked to clarify the role that biological (e.g., hormone levels, extent of physical damage), psychological (e.g., beliefs, coping responses), and social factors (e.g., social support) have in influencing the experience and impact of pain.[9] The findings from this research has informed the development of pain management multidisciplinary treatment programs, including psychosocial interventions, which are more effective in managing chronic pain than treatments that focus only on biomedical factors.[9–11]

More recently, researchers have become interested in understanding the association between pain, adjustment to pain, spirituality (and spiritual practices), and religiosity (and religious practices). Taken together, findings regarding the benefits of spiritual/religiosity in people with pain are encouraging. A recent systematic review summarized the evidence relative to the extent to which measures of spiritual/religiosity are associated with pain-related outcomes, pain-related beliefs, and pain coping

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responses in people with chronic pain.[12] The results from this review suggested that, regardless of the great variety of measures used to assess spirituality/religiosity and pain-related outcomes which limit the drawing of strong conclusions, a number of religiosity/spirituality domains are associated with lower pain (intensity and interference) and with better psychological function in people with chronic pain.

The findings from the body of research on the effects of specific spiritual and religious practices suggest that at least some of these practices may be useful resources in the context of pain care. For example, preliminary findings indicate that people who are more engaged with meaningful spiritual and religious practices – such as spiritual meditation and prayer – can tolerate pain better,[13,14] i.e., the maximum intensity of a pain-producing stimulus that a person is willing to accept in a particular situation is higher. [15] Prayer – defined as a form of spiritual communication between a human being with a deity (e.g., God) or with an object of prayer –[16] is among the spiritual and religious practices that have the potential to explain the beneficial effects of spirituality/religiosity on pain.

Prayer is among the most common and valued practices of individuals of both Western and Eastern countries of different religious denominations, even if they are not actively involved members of a religious tradition. [14] Prayer is a complex and multidimensional activity. Although there is a general consensus about its definition, there is not a dominant theoretical model or consistent approach to classify or measure different types of prayer.[17,18] In fact, previous research on the effects of prayer-based intervention for pain often fail to provide a clear definition of what is meant by "prayer" in any one study. These studies show a high degree of theoretical and methodological heterogeneity relative to the taxonomy of prayer employed, and its measurement.[17–19] While this heterogeneity makes comparisons across studies difficult, it also may

account, at least in part, for the inconsistent findings regarding the effects of prayer-based interventions on health- and pain-related outcomes, given the possibility that different types of prayer may have different effects on these outcomes.

Prayer may be *private* (or individual, i.e., an individual prays alone) or *communal* (i.e., an individual prays with another person or group of people), with respect to the number of people engaged in the prayer. [16,20] With respect to content, prayer can be of *petition* (or *supplication*, i.e., a request to the deity's or object of prayer's intervention towards oneself or towards another person or group of people), *thanksgiving* (i.e., expressing gratefulness), *adoration* (i.e., worshiping or praising the deity or object of prayer), *confession* (i.e., recognition of sins, misdeeds or faults), *reception* (i.e., waiting passively for the deity's or object of prayer's inspiration, wisdom and understanding), *meditation* (i.e., reflecting about a sacred text or a religious topic), or *ritualistic* (i.e., reading or reciting by memory a prayer or religious text).[17,20,21] Prayer can also be *participatory* or *distant intercessory*, with respect to the targeted beneficiary of the prayer. In *participatory* prayer, the targeted beneficiary of the prayer is one or more of the people doing the praying. In *distant* prayer, however, the targeted beneficiary of the praying is someone other than the person or people doing the praying. Of note, it is not the number of individuals involved in the praying (i.e., just one individual or more than one individual) that determines the content or beneficiary of the prayer. *Participatory* prayer, for example, may be *private* (or individual) or *communal* (i.e., dyadic or group-prayer) where the beneficiary of the prayer is/are either the individual or the people praying together. *Distant petition* prayer, either private or communal, where the beneficiary of the prayer is/are someone other than the person or people doing the praying, is often called *distant intercessory* prayer.

While previous research suggests the effects of *distant intercessory* prayer on health- and pain-related outcomes are weak,[18,22,23] previous findings relative to *participatory* prayer (i.e., i.e., individual, dyadic, or group-based prayer where the targeted beneficiary of the prayer is one or more of the people doing the praying) generally support stronger beneficial effects, and support its potential usefulness for patient care in the context of pain. For example, one study [24] with Muslim women who underwent a caesarean section showed that compared to a no treatment control, women randomized to an Islamic prayer condition showed significantly lower pain intensity scores, i.e., the magnitude of the experienced pain,[25] at three hours and six hours after the intervention.[24] Another study with patients who underwent gastrointestinal elective surgery, suggested that prayer as an adjunctive intervention to the use of analgesic medication contributed to reduced post operatory pain intensity to a greater extent than the use of analgesic medication alone.[26] Similar results were also found by Rahman and colleagues when comparing the post operatory pain intensity decrease of patients submitted to an emergency appendectomy in the prayer condition, to that observed in patients in the music-therapy and in the control conditions.[27] Although these and other studies suggest that spiritual/religiosity and spiritual/religious practices contribute to reduce self-reported pain-related distress in people with pain,[12] the results of studies focusing the effects of prayer-based interventions on objective stress responses are inconsistent; that is, a beneficial effect of prayer on self-reported pain-related stress and on objective stress responses (e.g., respiration and heart rate) is not always found.[24,28]

A recent systematic review summarized the evidence relative to the benefits of active personal *participatory* prayer as an intervention implemented on-site for pain.[19] Results from this review suggested that prayer appears to be useful in

reducing pain intensity and increasing pain tolerance, especially among religious people undergoing surgery or a painful procedure and engaging in active prayer to a deity.[19] However, this systematic review was limited to studies published in English between the years 2000 and 2019, in peer-reviewed journals indexed in one or more of only four databases (PubMed, Web of Science, PsychINFO, and Scopus). Thus, this review might have missed an important number of studies that: (a) were published in languages other than English, (b) were published before the year 2000, and (c) were published in journals indexed in other relevant databases or present in the grey literature. This systematic review was also restricted to the effects of *private* (*individual*) prayer on pain, thus excluding studies evaluating the effects of *communal* (either dyadic or group) prayer, which is a religious practice of many religions. Furthermore, this previous review did not provide: (a) a qualitative review of the findings based on a standardized and well-established method for qualitative synthesis of the findings; and (b) did not conduct a quality assessment of the studies included in the review. A comprehensive systematic review focusing the effects of both *private* (*individual*) and *communal* (dyadic and group) prayer on acute and chronic pain related-outcomes, including studies published both in peer-reviewed indexed journals and on grey literature regardless of the date of publication, using a standardized method for performing qualitative data analysis and metanalysis, and assessing both included studies characteristics and their methodological quality, would be useful for a more comprehensive and complete understanding of the usefulness of prayer as a pain management intervention. Hence, the present study aims to perform such a systematic review.

Study aims

The planned systematic review seeks to summarize and integrate the existing findings from randomized controlled trials assessing the effects of *participatory* prayer-

based interventions (including *private* and *communal*-based *participatory* prayer) on pain-related outcomes (i.e., pain intensity, pain tolerance, and pain-related stress). Specifically, this systematic review seeks to examine the extent to which prayer affects pain-related outcomes in adults with acute or chronic pain while including studies published in English, German, French, Spanish, Italian or Portuguese (i.e., the languages understood by one or more authors on the team), searched in nine databases (including one grey literature database), performing qualitative data analysis and studies' methodological quality assessment using standardized and well-established methods, and making recommendations for future research in this area. Given the subjective nature of pain's experience, this systematic review will be limited to studies that include a self-report of pain as a primary outcome.

METHOD

Review protocol and registration

The present systematic review will be conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement for systematic reviews and meta-analysis protocols.[29] Likewise, the PRISMA-P checklist was use to guide the protocol of this systematic review.[30] This review has been registered in the PROSPERO registry (PROSPERO 2020 CRD42020221733).

Review participants

Participants will be adult individuals experiencing pain, regardless of the cause of pain, pain duration, or pain location.

Eligibility criteria

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Articles will be included in this systematic review if they: (1) studied participants who are adults (i.e., 18 years old or older); (2) studied participants with pain, regardless of its duration or etiology; (3) included at least one quantitative measure of self-reported pain intensity, pain tolerance, and/or pain-related stress; (4) assessed the effects of *private* (*individual*) or *communal* (dyadic or group) *participatory* prayer-based interventions; (5) compared the intervention group to a control group (e.g., alternative treatment(s), treatment as usual, or both); (6) used a randomized controlled trial design; (7) were published in English, German, French, Spanish, Italian or Portuguese. At least two searches will be conducted; one for the first draft of the review paper, and a second search to update the review paper just prior to submission. While including only studies written in these languages might be viewed as restrictive, the authors are only proficient in these languages and do not have the resources to include articles written in other languages. Papers reporting the results of qualitative, cohort, case-control, and cross-sectional studies and papers focusing on spiritual practices, Complementary and Integrative Health interventions other than prayer (e.g., meditation, Reiki, and yoga), or *distant intercessory* prayer will be excluded from the systematic review.

Patient and public involvement

No patient is involved in either the design or planning phase of this study.

Search strategy

The systematic review protocol specifies and documents the search strategy, methods, and inclusion/exclusion criteria of the articles in advance. Search strategies to use to identify relevant publications will be customized to suit each database. Search terms to use in each database to search for publications regarding the previously mentioned pain-related outcomes will be (1) Religion OR Prayer AND (2) Pain AND

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(3) Pain intensity OR Pain tolerance OR Stress. The search terms will be used in various combinations, considering synonyms and MESH entries (*cf*. Supplementary file 1).

We will search 9 databases for the relevant publications: Web of Science Core Collection (Clarivate Analytics), MEDLINE (via Web of Science, Clarivate Analytics), SCIELO Citation Index (via Web of Science, Clarivate Analytics), PUBMED, Cochrane Central Register of Controlled Clinical Trial (via OvidSP), PsycINFO, Scopus, LILACS, and Open-SIGLE. Unpublished literature will also be searched in clinical trial registry platforms, such as ClinicalTrials.gov and ISRCTN registry. Then, to identify additional articles not identified during the database search, references list of articles included in the review and review articles found by the search will be hand searched.

Co-Primary outcomes

Studies should use at least one quantitative measure of pain intensity, pain tolerance, or pain-related stress to be included in the systematic review. The following are the commonly used measures assessing these variables.

Among the most common measures of pain intensity used by researchers and clinicians are the Visual Analogue Scale (VAS [31]) and related versions of this scale (e.g., mechanical VAS, digital VAS). Other commonly used measures of pain intensity [32] are the 0-10 Numerical Rating Scale (NRS), the Verbal Rating Scale (VRS), and the Faces Pain Scale-Revised (FPS-R [33]). Each of these measures is adequately valid and reliable as a measure of pain intensity, and patients in Western countries tend to prefer VRS and NRS over VAS.[32] Pain tolerance is commonly operationalized as the length (in seconds) that an individual is willing to experience an aversive stimulation.[34–36]

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Finally, perceived stress is usually assessed with the: (a) Perceived Stress Scale (PSS [37]), a 10-item scale that asks about people's feelings and thoughts regarding the passing month, on a scale from 0 (never) to 4 (very often); (b) stress scale of Depression, Anxiety, and Stress Scale (DASS [38]), a 14-item scale where subjects are asked to use 4-point severity/frequency scales to rate the extent to which they have experienced each state over the past week; and (c) 21-item version (DASS-21), which consist of a 7-item scale similar to the previous. In contrast, objective stress is usually measured through salivary cortisol, heart rate variability, or blood pressure measurement and analysis.[39–41]

Study selection, data management and data extraction

First, all references identified during the database search will be inserted into Zotero Software.[42] Second, cross-references and duplicates will be deleted. Then, two independent reviewers (MJar and IG) will screen the title, abstract, and keywords of the identified studies for eligibility screening based on the eligibility criteria listed above. Full texts of the manuscripts meeting the eligibility criteria and those with an uncertainty of inclusion from the abstracts will be independently read by the same reviewers to confirm inclusion or exclusion. Discrepancies found will be settled during a consensus meeting. In cases in which consensus is not achieved, a third reviewer will be consulted (AFV). The studies' inclusion or exclusion will be recorded on a self-designed data extraction form in Microsoft Excel 10.

Third, detailed data from the studies meeting the inclusion criteria will be extracted on a Microsoft Excel extraction sheet by two authors (MJar and IG). Data to be extracted include are: (1) authors and year of publication; (2) country of study; (3) study participant (type of pain – chronic or acute – and pain etiology); (4) sample size; (5) study participants age (mean, standard deviation); (6) percentage of female

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participants; (7) site of pain; (8) study participants' religious denomination and self-reported spirituality (e.g., self-proclaimed as "spiritual"); (9) study participants' type of religious practice (religious and a practitioner, religious but not a practitioner); (10) study participants' attitudes toward the religion (negative attitude towards deity vs. positive attitude towards deity); (11) length and intensity (frequency and length of sessions) of prayer-based intervention; (12) type of prayer (*private* or *communal* [i.e., dyadic or group] *participatory* prayer); (13) content of prayer (petition, thanksgiving, adoration, confession, reception, meditative, or ritualistic); (14) level of engagement and emotional involvement with the prayer-based intervention; (15) type and characteristics of the comparator or control condition; (16) self-reported pain intensity; (17) pain tolerance; and (18) pain-related stress. Data (mean, standard deviation, or effect sizes) on the primary outcomes will be extracted. Whenever possible, the rates of meaningful reductions in pain intensity – i.e., from responder analyses – will also be extracted; more specifically, the rates by which different groups obtained a 30% and a 50% reduction in pain intensity. Any discrepancies in the data collected will be resolved by discussion and consensus. In the event that consensus is not achieved, a third author (AFV) will be consulted.

Risk of bias assessment

The risk of bias of the studies included in the present systematic review will be evaluated based on the CONSORT (Consolidated Standards of Reporting Trials) statement. Risk of bias will be assessed by two independent reviewers (MJar and IG) and discrepancies will be settled during a consensus meeting. If a consensus is not reached, a third reviewer (AFV) will be consulted.

Data analysis plan

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Qualitative synthesis. A qualitative synthesis of the body of research will be conducted,[43] describing the methodological characteristics of the studies included in this systematic review, their strengths and limitations, and each study's results regarding the effects of *private* or *communal participatory* prayer in reducing pain intensity and pain-related stress and in increasing pain tolerance in adults experiencing pain, compared to treatment as usual or no treatment. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be summarized and examined separately. To summarize the evidence, we will implement a narrative summary synthesis method based on the framework developed by the UK Economic and Social Research Council to the conduct of narrative synthesis in systematic reviews,[44] modified to fit our purposes. Following the referred guidance, the synthesis process will involve three elements: (1) developing a preliminary synthesis of findings of included studies regarding the two topics; (2) exploring relationships in the data; and (3) assessing the robustness of the synthesis.

Meta-analysis. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be analyzed separately. If at least two included articles present data on a single primary outcome, then meta-analysis will be performed.[45] End-point scores will be computed as expressed mean differences or standardized mean difference (SMD) for each study for primary outcomes, with associated 95% confidence intervals. Data extracted in the articles will be transformed and presented in Cohen's *d* effect size with 95% confidence interval (*CI*) when possible using the Campbell Collaboration online calculator [46]. Cohen's *d* effect size thresholds to be used are: small (0.20), medium (0.50), large (0.80), and very large (1.30).[47]

Assessment of reporting biases

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We will use funnel plots to evaluate the potential reporting bias and small-study effects when there are more than 10 studies in the meta-analysis. Quantification of publication bias will be done using Egger's regression test method.[48]

Confidence in cumulative evidence

Included studies will be critically appraised and synthesized in terms of methodological quality along with the relevance, strength and limitations of the evidence presented. The similarities and differences between the studies (study characteristics, design and execution) will be described and their potential impact on study outcome will be discussed. The strength of the body of evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

DISCUSSION

The present systematic review will provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain. This allows for a more considerate and insightful choice when advising pain patients with effective coping strategies.

ETHICS AND DISSEMINATION

Ethics approval is not required for this review because it relies on secondary data. Any modification to the systematic review protocol will be submitted to review and approval of PROSPERO registry and described in the final report of the systematic review. The systematic review findings will be published in a peer-reviewed indexed journal and presented at a pain international scientific meeting.

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AUTHORS' CONTRIBUTIONS

AFV conceived the study idea, and together with MJar, MJ and MD developed the design of the protocol. IQG, FP, RC, JPR and MJ substantially contributed to the review and refinement of the review design and protocol. AFV and IQG planned the data analysis. AFV and MJar wrote the first draft of the manuscript. All authors critically reviewed the manuscript and approved the final version. AFV, MJar and IQG will

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conduct the review procedures. AFV is the principal investigator, obtained funding and is the guarantor of the review.

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COMPETING INTERESTS

None to declare.

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Appendix 1. Search terms, alternative search terms, and search strategy for Medline via Web of Science

	PAIN	PRAYER-BASED INTERVENTIONS	PAIN-RELATED OUTCOMES
MeSH heading	- Pain - Chronic pain - Acute pain	- Religion - Spiritual therapies	- Pain intensity - Pain tolerance - Stress
Title, abstract, and keywords	- Pain	- Religion - Pray* - Religious* - Spiritual therapies - Noetic therapies	- Pain intensity - Pain severity - Pain tolerance - Stress*

To identify articles related to prayer-based interventions

Pain

AND

Religion OR Pray* OR Religious* OR Spiritual therapies OR Noetic therapies

AND

Pain intensity OR Pain severity OR Pain tolerance OR Stress*

Limits: Adults >18 years; Humans; Randomized Control Trial

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BMJ Open

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Primary Subject Heading:	Complementary medicine
Secondary Subject Heading:	Anaesthesia, Research methods
Keywords:	PAIN MANAGEMENT, COMPLEMENTARY MEDICINE, STATISTICS & RESEARCH METHODS

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Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials

Alexandra Ferreira-Valente[1,2], Margarida Jarego[1], Inês Queiroz-Garcia[1], Filipa Pimenta[1], Rui Miguel Costa[1], Melissa A. Day[2,3], José Pais-Ribeiro[1], & Mark P. Jensen[2]

[1] William James Center for Research, ISPA – University Institute, Lisbon, Portugal

[2] University of Washington, Department of Rehabilitation Medicine, Seattle, WA, USA

[3] University of Queensland, School of Psychology, Faculty of Health & Behavioral Sciences, St. Lucia Campus, Brisbane, QLD 4072, Australia

Corresponding Author:

Alexandra Ferreira-Valente, PhD

William James Center for Research, ISPA – University Institute, Rua Jardim do Tabaco, n.º 34, 1149-041 Lisbon, Portugal

Tel.: +351 969082988

Email: mafvalente@gmail.com

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ABSTRACT

Background: Pain is a universal experience and the most common reason for seeking health care. Inadequate pain management negatively impacts numerous aspects of patient health. Multidisciplinary treatment programs, including psychosocial interventions, are more useful for pain management than purely biomedical treatment alone. Recently, researchers showed increasing interest in understanding the role of spirituality/religiosity and spiritual/religious practices on pain experience, with engagement in religious practices, such as prayer, showing to positively impact pain experience in religious individuals. This systematic review will seek to summarize and integrate the existing findings from randomized controlled trials assessing the effects of prayer and prayer-based interventions on pain experience.

Methods: The systematic review procedures and its report will follow the PRISMA statement checklist. Electronic searches in 9 databases (Web of Science Core Collection, MEDLINE, SCIELO Citation Index, PUBMED, Cochrane Central Register of Controlled Clinical Trial, PsycINFO, Scopus, LILACS, and Open-SIGLE) will be performed to identify randomized controlled trials of prayer-based interventions. Two independent researchers will assess studies for inclusion and extract data from each paper. The risk of bias assessment will be assessed independently by two reviewers based on the CONSORT statement. Qualitative synthesis of the body of research will be conducted using a narrative summary synthesis method. Meta-analysis will be limited to studies reporting on the same primary outcome.

Discussion: Findings will be useful to (1) understand the condition of our knowledge in this field; and (2) provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain.

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Systematic review registration: This protocol was submitted for registration at the International Prospective Register for Systematic Reviews (PROSPERO) with the reference CRD42020221733.

Keywords: Pain; Prayer; Systematic Review; Protocol

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is comprehensive systematic review focusing the effects of prayer (despite the type of prayer) on pain related-outcomes (despite the type of pain), including studies published in peer-reviewed indexed journals and on grey literature regardless of the date of publication
- PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) checklist guidelines will be followed to conduct this review, and PRISMA-P checklist was followed to report the protocol of this systematic review, to ensure quality in all aspects of study planning, execution and reporting.
- Risk of bias assessment, assessment of reporting and publication bias, and evaluation of the confidence in cumulative evidence will be performed, respectively, based on CONSORT (Consolidated Standards of Reporting Trials) statement, Egger's regression test method, and on the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.
- Main strengths of the study are its comprehensiveness, including studies and grey literature published in five languages despite the date of publication, and the use of well-established and standardized methods to perform qualitative synthesis, meta-analysis, risk of bias assessment and meta-bias assessment.
- We may not be able to conduct the anticipated meta-analysis due to methodological heterogeneity between the included studies and between the primary outcomes' measures used in the different studies.

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INTRODUCTION

Pain – defined by the International Association for the Study of Pain as an "unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage" –[1] is a universal experience. [2] Despite its unpleasantness, pain – except when associated with many chronic pain conditions – usually has an adaptive purpose. It is a sign that physical damage might be occurring that threatens the organism's well-being.[3] Pain is the most common reason for seeking health care,[4] and inadequate management of acute pain negatively impacts numerous aspects of patient health. This includes reduced quality of life, impaired sleep, impaired physical function, and increased risk of developing chronic pain.[5] In short, pain remains a significant health issue, and its management a priority worldwide.[5–8]

Because pain is a multidimensional and subjective experience, researchers and clinicians have worked to clarify the role that biological (e.g., hormone levels, extent of physical damage), psychological (e.g., beliefs, coping responses), and social factors (e.g., social support) have in influencing the experience and impact of pain.[9] The findings from this research has informed the development of pain management multidisciplinary treatment programs, including psychosocial interventions, which are more effective in managing chronic pain than treatments that focus only on biomedical factors.[9–11]

More recently, researchers have become interested in understanding the association between pain, adjustment to pain, spirituality (and spiritual practices), and religiosity (and religious practices). Taken together, findings regarding the benefits of spiritual/religiosity in people with pain are encouraging. A recent systematic review summarized the evidence relative to the extent to which measures of spiritual/religiosity are associated with pain-related outcomes, pain-related beliefs, and pain coping

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responses in people with chronic pain.[12] The results from this review suggested that, regardless of the great variety of measures used to assess spirituality/religiosity and pain-related outcomes which limit the drawing of strong conclusions, a number of religiosity/spirituality domains are associated with lower pain (intensity and interference) and with better psychological function in people with chronic pain.

The findings from the body of research on the effects of specific spiritual and religious practices suggest that at least some of these practices may be useful resources in the context of pain care. For example, preliminary findings indicate that people who are more engaged with meaningful spiritual and religious practices – such as spiritual meditation and prayer – can tolerate pain better,[13,14] i.e., the maximum intensity of a pain-producing stimulus that a person is willing to accept in a particular situation is higher. [15] Prayer – defined as a form of spiritual communication between a human being with a deity (e.g., God) or with an object of prayer –[16] is among the spiritual and religious practices that have the potential to explain the beneficial effects of spirituality/religiosity on pain.

Prayer is among the most common and valued practices of individuals of both Western and Eastern countries of different religious denominations, even if they are not actively involved members of a religious tradition. [14] Prayer is a complex and multidimensional activity. Although there is a general consensus about its definition, there is not a dominant theoretical model or consistent approach to classify or measure different types of prayer.[17,18] In fact, previous research on the effects of prayer-based intervention for pain often fail to provide a clear definition of what is meant by "prayer" in any one study. These studies show a high degree of theoretical and methodological heterogeneity relative to the taxonomy of prayer employed, and its measurement.[17–19] While this heterogeneity makes comparisons across studies difficult, it also may

account, at least in part, for the inconsistent findings regarding the effects of prayer-based interventions on health- and pain-related outcomes, given the possibility that different types of prayer may have different effects on these outcomes.

Prayer may be *private* (or individual, i.e., an individual prays alone) or *communal* (i.e., an individual prays with another person or group of people), with respect to the number of people engaged in the prayer. [16,20] With respect to content, prayer can be of *petition* (or *supplication*, i.e., a request to the deity's or object of prayer's intervention towards oneself or towards another person or group of people), *thanksgiving* (i.e., expressing gratefulness), *adoration* (i.e., worshiping or praising the deity or object of prayer), *confession* (i.e., recognition of sins, misdeeds or faults), *reception* (i.e., waiting passively for the deity's or object of prayer's inspiration, wisdom and understanding), *meditation* (i.e., reflecting about a sacred text or a religious topic), or *ritualistic* (i.e., reading or reciting by memory a prayer or religious text).[17,20,21] Prayer can also be *participatory* or *distant intercessory*, with respect to the targeted beneficiary of the prayer. In *participatory* prayer, the targeted beneficiary of the prayer is one or more of the people doing the praying. In *distant* prayer, however, the targeted beneficiary of the praying is someone other than the person or people doing the praying. Of note, it is not the number of individuals involved in the praying (i.e., just one individual or more than one individual) that determines the content or beneficiary of the prayer. *Participatory* prayer, for example, may be *private* (or individual) or *communal* (i.e., dyadic or group-prayer) where the beneficiary of the prayer is/are either the individual or the people praying together. *Distant petition* prayer, either private or communal, where the beneficiary of the prayer is/are someone other than the person or people doing the praying, is often called *distant intercessory* prayer.

While previous research suggests the effects of *distant intercessory* prayer on health- and pain-related outcomes are weak,[18,22,23] previous findings relative to *participatory* prayer (i.e., i.e., individual, dyadic, or group-based prayer where the targeted beneficiary of the prayer is one or more of the people doing the praying) generally support stronger beneficial effects, and support its potential usefulness for patient care in the context of pain. For example, one study [24] with Muslim women who underwent a caesarean section showed that compared to a no treatment control, women randomized to an Islamic prayer condition showed significantly lower pain intensity scores, i.e., the magnitude of the experienced pain,[25] at three hours and six hours after the intervention.[24] Another study with patients who underwent gastrointestinal elective surgery, suggested that prayer as an adjunctive intervention to the use of analgesic medication contributed to reduced post operatory pain intensity to a greater extent than the use of analgesic medication alone.[26] Similar results were also found by Rahman and colleagues when comparing the post operatory pain intensity decrease of patients submitted to an emergency appendectomy in the prayer condition, to that observed in patients in the music-therapy and in the control conditions.[27] Although these and other studies suggest that spiritual/religiosity and spiritual/religious practices contribute to reduce self-reported pain-related distress in people with pain,[12] the results of studies focusing the effects of prayer-based interventions on objective stress responses are inconsistent; that is, a beneficial effect of prayer on self-reported pain-related stress and on objective stress responses (e.g., respiration and heart rate) is not always found.[24,28]

A recent systematic review summarized the evidence relative to the benefits of active personal *participatory* prayer as an intervention implemented on-site for pain.[19] Results from this review suggested that prayer appears to be useful in

reducing pain intensity and increasing pain tolerance, especially among religious people undergoing surgery or a painful procedure and engaging in active prayer to a deity.[19] However, this systematic review was limited to studies published in English between the years 2000 and 2019, in peer-reviewed journals indexed in one or more of only four databases (PubMed, Web of Science, PsychINFO, and Scopus). Thus, this review might have missed an important number of studies that: (a) were published in languages other than English, (b) were published before the year 2000, and (c) were published in journals indexed in other relevant databases or present in the grey literature. This systematic review was also restricted to the effects of *private* (*individual*) prayer on pain, thus excluding studies evaluating the effects of *communal* (either dyadic or group) prayer, which is a religious practice of many religions. Furthermore, this previous review did not provide: (a) a qualitative review of the findings based on a standardized and well-established method for qualitative synthesis of the findings; and (b) did not conduct a quality assessment of the studies included in the review. A comprehensive systematic review focusing the effects of both *private* (*individual*) and *communal* (dyadic and group) prayer on acute and chronic pain related-outcomes, including studies published both in peer-reviewed indexed journals and on grey literature regardless of the date of publication, using a standardized method for performing qualitative data analysis and metanalysis, and assessing both included studies characteristics and their methodological quality, would be useful for a more comprehensive and complete understanding of the usefulness of prayer as a pain management intervention. Hence, the present study aims to perform such a systematic review.

Study aims

The planned systematic review seeks to summarize and integrate the existing findings from randomized controlled trials assessing the effects of *participatory* prayer-

based interventions (including *private* and *communal*-based *participatory* prayer) on pain-related outcomes (i.e., pain intensity, pain tolerance, and pain-related stress). Specifically, this systematic review seeks to examine the extent to which prayer affects pain-related outcomes in adults with acute or chronic pain while including studies published in English, German, French, Spanish, Italian or Portuguese (i.e., the languages understood by one or more authors on the team), searched in nine databases (including one grey literature database), performing qualitative data analysis and studies' methodological quality assessment using standardized and well-established methods, and making recommendations for future research in this area. Given the subjective nature of pain's experience, this systematic review will be limited to studies that include a self-report of pain as a primary outcome.

METHOD

Review protocol and registration

The present systematic review will be conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement for systematic reviews and meta-analysis protocols.[29] Likewise, the PRISMA-P checklist was will be used to guide the protocol of this systematic review.[30] This review has been registered in the PROSPERO registry (PROSPERO 2020 CRD42020221733).

Review participants

Participants will be adult individuals experiencing pain, regardless of the cause of pain, pain duration, or pain location.

Eligibility criteria

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Articles will be included in this systematic review if they: (1) studied participants who are adults (i.e., 18 years old or older); (2) studied participants with pain, regardless of its duration or etiology; (3) included at least one quantitative measure of self-reported pain intensity, pain tolerance, and/or pain-related stress; (4) assessed the effects of *private* (*individual*) or *communal* (dyadic or group) *participatory* prayer-based interventions; (5) compared the intervention group to a control group (e.g., alternative treatment(s), treatment as usual, or both); (6) used a randomized controlled trial design; (7) were published in English, German, French, Spanish, Italian or Portuguese. At least two searches will be conducted; one for the first draft of the review paper, and a second search to update the review paper just prior to submission. While including only studies written in these languages might be viewed as restrictive, the authors are only proficient in these languages and do not have the resources to include articles written in other languages. Papers reporting the results of qualitative, cohort, case-control, and cross-sectional studies and papers focusing on spiritual practices, Complementary and Integrative Health interventions other than prayer (e.g., meditation, Reiki, and yoga), or *distant intercessory* prayer will be excluded from the systematic review.

Patient and public involvement

No patient is involved in either the design or planning phase of this study.

Search strategy

The systematic review protocol specifies and documents the search strategy, methods, and inclusion/exclusion criteria of the articles in advance. The search strategies that will be used to identify relevant publications will be customized to suit each database. The search terms will be (1) Religion OR Prayer AND (2) Pain AND (3)

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Pain intensity OR Pain tolerance OR Stress. The search terms will be used in various combinations, considering synonyms and MESH entries (*cf*. Supplementary file 1).

We will search 9 databases for the relevant publications: Web of Science Core Collection (Clarivate Analytics), MEDLINE (via Web of Science, Clarivate Analytics), SCIELO Citation Index (via Web of Science, Clarivate Analytics), PUBMED, Cochrane Central Register of Controlled Clinical Trial (via OvidSP), PsycINFO, Scopus, LILACS, and Open-SIGLE. Unpublished literature will also be searched in clinical trial registry platforms, such as ClinicalTrials.gov and ISRCTN registry. Then, to identify additional articles not identified during the database search, references list of articles included in the review and review articles found by the search will be hand searched.

Co-Primary outcomes

Studies will be included if they included at least one quantitative measure of pain intensity, pain tolerance, or pain-related stress to be included in the systematic review. The following are the commonly used measures assessing these variables.

Among the most common measures of pain intensity used by researchers and clinicians are the Visual Analogue Scale (VAS [31]) and related versions of this scale (e.g., mechanical VAS, digital VAS). Other commonly used measures of pain intensity [32] are the 0-10 Numerical Rating Scale (NRS), the Verbal Rating Scale (VRS), and the Faces Pain Scale-Revised (FPS-R [33]). Each of these measures is adequately valid and reliable as a measure of pain intensity, and patients in Western countries tend to prefer VRS and NRS over VAS.[32] Pain tolerance is commonly operationalized as the length (in seconds) that an individual is willing to experience an aversive stimulation.[34–36]

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Finally, perceived stress is usually assessed with the: (a) Perceived Stress Scale (PSS [37]), a 10-item scale that asks about people's feelings and thoughts regarding the passing month, on a scale from 0 (never) to 4 (very often); (b) stress scale of Depression, Anxiety, and Stress Scale (DASS [38]), a 14-item scale where subjects are asked to use 4-point severity/frequency scales to rate the extent to which they have experienced each state over the past week; and (c) 21-item version (DASS-21), which consist of a 7-item scale similar to the previous. In contrast, objective stress is usually measured through salivary cortisol, heart rate variability, or blood pressure measurement and analysis.[39–41]

Study selection, data management and data extraction

First, all references identified during the database search will be inserted into Zotero Software.[42] Second, cross-references and duplicates will be deleted. Then, two independent reviewers (MJar and IG) will screen the title, abstract, and keywords of the identified studies for eligibility screening based on the eligibility criteria listed above. Full texts of the manuscripts meeting the eligibility criteria and those with an uncertainty of inclusion from the abstracts will be independently read by the same reviewers to confirm inclusion or exclusion. Discrepancies found will be settled during a consensus meeting. In cases in which consensus is not achieved, a third reviewer will be consulted (AFV). The studies' inclusion or exclusion will be recorded on a self-designed data extraction form in Microsoft Excel 10.

Third, detailed data from the studies meeting the inclusion criteria will be extracted on a Microsoft Excel extraction sheet by two authors (MJar and IG). Data to be extracted include are: (1) authors and year of publication; (2) country of study; (3) study participant (type of pain – chronic or acute – and pain etiology); (4) sample size; (5) study participants age (mean, standard deviation); (6) percentage of female

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participants; (7) site of pain; (8) study participants' religious denomination and self-reported spirituality (e.g., self-proclaimed as "spiritual"); (9) study participants' type of religious practice (religious and a practitioner, religious but not a practitioner); (10) study participants' attitudes toward the religion (negative attitude towards deity vs. positive attitude towards deity); (11) length and intensity (frequency and length of sessions) of prayer-based intervention; (12) type of prayer (*private* or *communal* [i.e., dyadic or group] *participatory* prayer); (13) content of prayer (petition, thanksgiving, adoration, confession, reception, meditative, or ritualistic); (14) level of engagement and emotional involvement with the prayer-based intervention; (15) type and characteristics of the comparator or control condition; (16) self-reported pain intensity; (17) pain tolerance; and (18) pain-related stress. Data (mean, standard deviation, or effect sizes) on the primary outcomes will be extracted. Whenever possible, the rates of meaningful reductions in pain intensity – i.e., from responder analyses – will also be extracted; more specifically, the rates by which different groups obtained a 30% and a 50% reduction in pain intensity. Any discrepancies in the data collected will be resolved by discussion and consensus. In the event that consensus is not achieved, a third author (AFV) will be consulted.

Risk of bias assessment

The risk of bias of the studies included in the present systematic review will be evaluated based on the CONSORT (Consolidated Standards of Reporting Trials) statement. Risk of bias will be assessed by two independent reviewers (MJar and IG) and discrepancies will be settled during a consensus meeting. If a consensus is not reached, a third reviewer (AFV) will be consulted.

Data analysis plan

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Qualitative synthesis. A qualitative synthesis of the body of research will be conducted,[43] describing the methodological characteristics of the studies included in this systematic review, their strengths and limitations, and each study's results regarding the effects of *private* or *communal participatory* prayer in reducing pain intensity and pain-related stress and in increasing pain tolerance in adults experiencing pain, compared to treatment as usual or no treatment. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be summarized and examined separately. To summarize the evidence, we will implement a narrative summary synthesis method based on the framework developed by the UK Economic and Social Research Council to the conduct of narrative synthesis in systematic reviews,[44] modified to fit our purposes. Following the referred guidance, the synthesis process will involve three elements: (1) developing a preliminary synthesis of findings of included studies regarding the two topics; (2) exploring relationships in the data; and (3) assessing the robustness of the synthesis.

Meta-analysis. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be analyzed separately. If at least two included articles present data on a single primary outcome, then meta-analysis will be performed.[45] End-point scores will be computed as expressed mean differences or standardized mean difference (SMD) for each study for primary outcomes, with associated 95% confidence intervals. Data extracted in the articles will be transformed and presented in Cohen's *d* effect size with 95% confidence interval (*CI*) when possible using the Campbell Collaboration online calculator [46]. The Cohen's *d* effect size thresholds we will use: small (0.20), medium (0.50), large (0.80), and very large (1.30).[47]

Assessment of reporting biases

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We will use funnel plots to evaluate the potential reporting bias and small-study effects when there are more than 10 studies in the meta-analysis. Quantification of publication bias will be done using Egger's regression test method.[48]

Confidence in cumulative evidence

Included studies will be critically appraised and synthesized in terms of methodological quality along with the relevance, strength and limitations of the evidence presented. The similarities and differences between the studies (study characteristics, design and execution) will be described and their potential impact on study outcome will be discussed. The strength of the body of evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

DISCUSSION

The present systematic review will provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain. This allows for a more considerate and insightful choice when advising pain patients with effective coping strategies.

ETHICS AND DISSEMINATION

Ethics approval is not required for this review because it relies on secondary data. Any modification to the systematic review protocol will be submitted to review and approval of PROSPERO registry and described in the final report of the systematic review. The systematic review findings will be published in a peer-reviewed indexed journal and presented at a pain international scientific meeting.

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Not applicable.

AUTHORS' CONTRIBUTIONS

AFV conceived the study idea, and together with MJar, MJ and MD developed the design of the protocol. IQG, FP, RC, JPR and MJ substantially contributed to the review and refinement of the review design and protocol. AFV and IQG planned the data analysis. AFV and MJar wrote the first draft of the manuscript. All authors critically reviewed the manuscript and approved the final version. AFV, MJar and IQG will

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conduct the review procedures. AFV is the principal investigator, obtained funding and is the guarantor of the review.

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COMPETING INTERESTS

None to declare.

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Appendix 1. Search terms, alternative search terms, and search strategy for Medline via Web of Science

	PAIN	PRAYER-BASED INTERVENTIONS	PAIN-RELATED OUTCOMES
MeSH heading	- Pain - Chronic pain - Acute pain	- Religion - Spiritual therapies	- Pain intensity - Pain tolerance - Stress
Title, abstract, and keywords	- Pain	- Religion - Pray* - Religious* - Spiritual therapies - Noetic therapies	- Pain intensity - Pain severity - Pain tolerance - Stress*

To identify articles related to prayer-based interventions

Pain

AND

Religion OR Pray* OR Religious* OR Spiritual therapies OR Noetic therapies

AND

Pain intensity OR Pain severity OR Pain tolerance OR Stress*

Limits: Adults >18 years; Humans; Randomized Control Trial

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BMJ Open

Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047580.R3
Article Type:	Protocol
Date Submitted by the Author:	16-Jun-2021
Complete List of Authors:	Ferreira-Valente, Alexandra; ISPA, William James Center for Research; University of Washington, Department of Rehabilitation Medicine Jarego, Margarida; Higher Institute of Applied Psychology William James Center for Research Queiroz-Garcia, Inês; Higher Institute of Applied Psychology William James Center for Research Pimenta, Filipa; ISPA, William James Center for Research Costa, Rui; ISPA, William James Center for Research Day, Melissa ; University of Queensland School of Psychology; University of Washington, Department of Rehabilitation Medicine Pais-Ribeiro, José; ISPA, William James Center for Research Jensen, Mark; University of Washington, Departement of Rehabilitation Medicine
Primary Subject Heading:	Complementary medicine
Secondary Subject Heading:	Anaesthesia, Research methods
Keywords:	PAIN MANAGEMENT, COMPLEMENTARY MEDICINE, STATISTICS & RESEARCH METHODS

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Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials

Alexandra Ferreira-Valente[1,2], Margarida Jarego[1], Inês Queiroz-Garcia[1], Filipa Pimenta[1], Rui Miguel Costa[1], Melissa A. Day[2,3], José Pais-Ribeiro[1], & Mark P. Jensen[2]

[1] William James Center for Research, ISPA – University Institute, Lisbon, Portugal

[2] University of Washington, Department of Rehabilitation Medicine, Seattle, WA, USA

[3] University of Queensland, School of Psychology, Faculty of Health & Behavioral Sciences, St. Lucia Campus, Brisbane, QLD 4072, Australia

Corresponding Author:

Alexandra Ferreira-Valente, PhD

William James Center for Research, ISPA – University Institute, Rua Jardim do Tabaco, n.º 34, 1149-041 Lisbon, Portugal

Tel.: +351 969082988

Email: mafvalente@gmail.com

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ABSTRACT

Background: Pain is a universal experience and the most common reason for seeking health care. Inadequate pain management negatively impacts numerous aspects of patient health. Multidisciplinary treatment programs, including psychosocial interventions, are more useful for pain management than purely biomedical treatment alone. Recently, researchers showed increasing interest in understanding the role of spirituality/religiosity and spiritual/religious practices on pain experience, with engagement in religious practices, such as prayer, showing to positively impact pain experience in religious individuals. This systematic review will seek to summarize and integrate the existing findings from randomized controlled trials assessing the effects of prayer and prayer-based interventions on pain experience.

Methods: The systematic review procedures and its report will follow the PRISMA statement checklist. Electronic searches in 9 databases (Web of Science Core Collection, MEDLINE, SCIELO Citation Index, PUBMED, Cochrane Central Register of Controlled Clinical Trial, PsycINFO, Scopus, LILACS, and Open-SIGLE) will be performed to identify randomized controlled trials of prayer-based interventions. Two independent researchers will assess studies for inclusion and extract data from each paper. The risk of bias assessment will be assessed independently by two reviewers based on the CONSORT statement. Qualitative synthesis of the body of research will be conducted using a narrative summary synthesis method. Meta-analysis will be limited to studies reporting on the same primary outcome. Formal searches are planned to start in June 2021. Final report is anticipated to be completed by September 2021.

Discussion: Findings will be useful to (1) understand the condition of our knowledge in this field; and (2) provide evidence for prayer effectiveness in reducing pain intensity

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and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain.

Systematic review registration: This protocol was submitted for registration at the International Prospective Register for Systematic Reviews (PROSPERO) with the reference CRD42020221733.

Keywords: Pain; Prayer; Systematic Review; Protocol

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is comprehensive systematic review focusing the effects of prayer (despite the type of prayer) on pain related-outcomes (despite the type of pain), including studies published in peer-reviewed indexed journals and on grey literature regardless of the date of publication
- PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) checklist guidelines will be followed to conduct this review, and PRISMA-P checklist was followed to report the protocol of this systematic review, to ensure quality in all aspects of study planning, execution and reporting.
- Risk of bias assessment, assessment of reporting and publication bias, and evaluation of the confidence in cumulative evidence will be performed, respectively, based on CONSORT (Consolidated Standards of Reporting Trials) statement, Egger's regression test method, and on the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.
- Main strengths of the study are its comprehensiveness, including studies and grey literature published in five languages despite the date of publication, and the use of well-established and standardized methods to perform qualitative synthesis, meta-analysis, risk of bias assessment and meta-bias assessment.
- We may not be able to conduct the anticipated meta-analysis due to methodological heterogeneity between the included studies and between the primary outcomes' measures used in the different studies.

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INTRODUCTION

Pain – defined by the International Association for the Study of Pain as an "unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage" –[1] is a universal experience. [2] Despite its unpleasantness, pain – except when associated with many chronic pain conditions – usually has an adaptive purpose. It is a sign that physical damage might be occurring that threatens the organism's well-being.[3] Pain is the most common reason for seeking health care,[4] and inadequate management of acute pain negatively impacts numerous aspects of patient health. This includes reduced quality of life, impaired sleep, impaired physical function, and increased risk of developing chronic pain.[5] In short, pain remains a significant health issue, and its management a priority worldwide.[5–8]

Because pain is a multidimensional and subjective experience, researchers and clinicians have worked to clarify the role that biological (e.g., hormone levels, extent of physical damage), psychological (e.g., beliefs, coping responses), and social factors (e.g., social support) have in influencing the experience and impact of pain.[9] The findings from this research has informed the development of pain management multidisciplinary treatment programs, including psychosocial interventions, which are more effective in managing chronic pain than treatments that focus only on biomedical factors.[9–11]

More recently, researchers have become interested in understanding the association between pain, adjustment to pain, spirituality (and spiritual practices), and religiosity (and religious practices). Taken together, findings regarding the benefits of spiritual/religiosity in people with pain are encouraging. A recent systematic review summarized the evidence relative to the extent to which measures of spiritual/religiosity are associated with pain-related outcomes, pain-related beliefs, and pain coping

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responses in people with chronic pain.[12] The results from this review suggested that, regardless of the great variety of measures used to assess spirituality/religiosity and pain-related outcomes which limit the drawing of strong conclusions, a number of religiosity/spirituality domains are associated with lower pain (intensity and interference) and with better psychological function in people with chronic pain.

The findings from the body of research on the effects of specific spiritual and religious practices suggest that at least some of these practices may be useful resources in the context of pain care. For example, preliminary findings indicate that people who are more engaged with meaningful spiritual and religious practices – such as spiritual meditation and prayer – can tolerate pain better,[13,14] i.e., the maximum intensity of a pain-producing stimulus that a person is willing to accept in a particular situation is higher. [15] Prayer – defined as a form of spiritual communication between a human being with a deity (e.g., God) or with an object of prayer –[16] is among the spiritual and religious practices that have the potential to explain the beneficial effects of spirituality/religiosity on pain.

Prayer is among the most common and valued practices of individuals of both Western and Eastern countries of different religious denominations, even if they are not actively involved members of a religious tradition. [14] Prayer is a complex and multidimensional activity. Although there is a general consensus about its definition, there is not a dominant theoretical model or consistent approach to classify or measure different types of prayer.[17,18] In fact, previous research on the effects of prayer-based intervention for pain often fail to provide a clear definition of what is meant by "prayer" in any one study. These studies show a high degree of theoretical and methodological heterogeneity relative to the taxonomy of prayer employed, and its measurement.[17–19] While this heterogeneity makes comparisons across studies difficult, it also may

account, at least in part, for the inconsistent findings regarding the effects of prayer-based interventions on health- and pain-related outcomes, given the possibility that different types of prayer may have different effects on these outcomes.

Prayer may be *private* (or individual, i.e., an individual prays alone) or *communal* (i.e., an individual prays with another person or group of people), with respect to the number of people engaged in the prayer. [16,20] With respect to content, prayer can be of *petition* (or *supplication*, i.e., a request to the deity's or object of prayer's intervention towards oneself or towards another person or group of people), *thanksgiving* (i.e., expressing gratefulness), *adoration* (i.e., worshiping or praising the deity or object of prayer), *confession* (i.e., recognition of sins, misdeeds or faults), *reception* (i.e., waiting passively for the deity's or object of prayer's inspiration, wisdom and understanding), *meditation* (i.e., reflecting about a sacred text or a religious topic), or *ritualistic* (i.e., reading or reciting by memory a prayer or religious text).[17,20,21] Prayer can also be *participatory* or *distant intercessory*, with respect to the targeted beneficiary of the prayer. In *participatory* prayer, the targeted beneficiary of the prayer is one or more of the people doing the praying. In *distant* prayer, however, the targeted beneficiary of the praying is someone other than the person or people doing the praying. Of note, it is not the number of individuals involved in the praying (i.e., just one individual or more than one individual) that determines the content or beneficiary of the prayer. *Participatory* prayer, for example, may be *private* (or individual) or *communal* (i.e., dyadic or group-prayer) where the beneficiary of the prayer is/are either the individual or the people praying together. *Distant petition* prayer, either private or communal, where the beneficiary of the prayer is/are someone other than the person or people doing the praying, is often called *distant intercessory* prayer.

While previous research suggests the effects of *distant intercessory* prayer on health- and pain-related outcomes are weak,[18,22,23] previous findings relative to *participatory* prayer (i.e., i.e., individual, dyadic, or group-based prayer where the targeted beneficiary of the prayer is one or more of the people doing the praying) generally support stronger beneficial effects, and support its potential usefulness for patient care in the context of pain. For example, one study [24] with Muslim women who underwent a caesarean section showed that compared to a no treatment control, women randomized to an Islamic prayer condition showed significantly lower pain intensity scores, i.e., the magnitude of the experienced pain,[25] at three hours and six hours after the intervention.[24] Another study with patients who underwent gastrointestinal elective surgery, suggested that prayer as an adjunctive intervention to the use of analgesic medication contributed to reduced post operatory pain intensity to a greater extent than the use of analgesic medication alone.[26] Similar results were also found by Rahman and colleagues when comparing the post operatory pain intensity decrease of patients submitted to an emergency appendectomy in the prayer condition, to that observed in patients in the music-therapy and in the control conditions.[27] Although these and other studies suggest that spiritual/religiosity and spiritual/religious practices contribute to reduce self-reported pain-related distress in people with pain,[12] the results of studies focusing the effects of prayer-based interventions on objective stress responses are inconsistent; that is, a beneficial effect of prayer on self-reported pain-related stress and on objective stress responses (e.g., respiration and heart rate) is not always found.[24,28]

A recent systematic review summarized the evidence relative to the benefits of active personal *participatory* prayer as an intervention implemented on-site for pain.[19] Results from this review suggested that prayer appears to be useful in

reducing pain intensity and increasing pain tolerance, especially among religious people undergoing surgery or a painful procedure and engaging in active prayer to a deity.[19] However, this systematic review was limited to studies published in English between the years 2000 and 2019, in peer-reviewed journals indexed in one or more of only four databases (PubMed, Web of Science, PsychINFO, and Scopus). Thus, this review might have missed an important number of studies that: (a) were published in languages other than English, (b) were published before the year 2000, and (c) were published in journals indexed in other relevant databases or present in the grey literature. This systematic review was also restricted to the effects of *private* (*individual*) prayer on pain, thus excluding studies evaluating the effects of *communal* (either dyadic or group) prayer, which is a religious practice of many religions. Furthermore, this previous review did not provide: (a) a qualitative review of the findings based on a standardized and well-established method for qualitative synthesis of the findings; and (b) did not conduct a quality assessment of the studies included in the review. A comprehensive systematic review focusing the effects of both *private* (*individual*) and *communal* (dyadic and group) prayer on acute and chronic pain related-outcomes, including studies published both in peer-reviewed indexed journals and on grey literature regardless of the date of publication, using a standardized method for performing qualitative data analysis and metanalysis, and assessing both included studies characteristics and their methodological quality, would be useful for a more comprehensive and complete understanding of the usefulness of prayer as a pain management intervention. Hence, the present study aims to perform such a systematic review.

Study aims

The planned systematic review seeks to summarize and integrate the existing findings from randomized controlled trials assessing the effects of *participatory* prayer-

based interventions (including *private* and *communal*-based *participatory* prayer) on pain-related outcomes (i.e., pain intensity, pain tolerance, and pain-related stress). Specifically, this systematic review seeks to examine the extent to which prayer affects pain-related outcomes in adults with acute or chronic pain while including studies published in English, German, French, Spanish, Italian or Portuguese (i.e., the languages understood by one or more authors on the team), searched in nine databases (including one grey literature database), performing qualitative data analysis and studies' methodological quality assessment using standardized and well-established methods, and making recommendations for future research in this area. Given the subjective nature of pain's experience, this systematic review will be limited to studies that include a self-report of pain as a primary outcome.

METHOD

Review protocol and registration

The present systematic review will be conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement for systematic reviews and meta-analysis protocols.[29] Likewise, the PRISMA-P checklist was will be used to guide the protocol of this systematic review.[30] This review has been registered in the PROSPERO registry (PROSPERO 2020 CRD42020221733).

Review participants

Participants will be adult individuals experiencing pain, regardless of the cause of pain, pain duration, or pain location.

Eligibility criteria

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Articles will be included in this systematic review if they: (1) studied participants who are adults (i.e., 18 years old or older); (2) studied participants with pain, regardless of its duration or etiology; (3) included at least one quantitative measure of self-reported pain intensity, pain tolerance, and/or pain-related stress; (4) assessed the effects of *private* (*individual*) or *communal* (dyadic or group) *participatory* prayer-based interventions; (5) compared the intervention group to a control group (e.g., alternative treatment(s), treatment as usual, or both); (6) used a randomized controlled trial design; (7) were published in English, German, French, Spanish, Italian or Portuguese. At least two searches will be conducted; one for the first draft of the review paper, and a second search to update the review paper just prior to submission. While including only studies written in these languages might be viewed as restrictive, the authors are only proficient in these languages and do not have the resources to include articles written in other languages. Papers reporting the results of qualitative, cohort, case-control, and cross-sectional studies and papers focusing on spiritual practices, Complementary and Integrative Health interventions other than prayer (e.g., meditation, Reiki, and yoga), or *distant intercessory* prayer will be excluded from the systematic review.

Patient and public involvement

No patient is involved in either the design or planning phase of this study.

Search strategy

The systematic review protocol specifies and documents the search strategy, methods, and inclusion/exclusion criteria of the articles in advance. The search strategies that will be used to identify relevant publications will be customized to suit each database. The search terms will be (1) Religion OR Prayer AND (2) Pain AND (3)

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Pain intensity OR Pain tolerance OR Stress. The search terms will be used in various combinations, considering synonyms and MESH entries (*cf*. Supplementary file 1).

We will search 9 databases for the relevant publications: Web of Science Core Collection (Clarivate Analytics), MEDLINE (via Web of Science, Clarivate Analytics), SCIELO Citation Index (via Web of Science, Clarivate Analytics), PUBMED, Cochrane Central Register of Controlled Clinical Trial (via OvidSP), PsycINFO, Scopus, LILACS, and Open-SIGLE. Unpublished literature will also be searched in clinical trial registry platforms, such as ClinicalTrials.gov and ISRCTN registry. Then, to identify additional articles not identified during the database search, references list of articles included in the review and review articles found by the search will be hand searched. Preliminary searches were planned to start in December 2020. Formal searches are planned to be completed in July 2021.

Co-Primary outcomes

Studies will be included if they included at least one quantitative measure of pain intensity, pain tolerance, or pain-related stress to be included in the systematic review. The following are the commonly used measures assessing these variables.

Among the most common measures of pain intensity used by researchers and clinicians are the Visual Analogue Scale (VAS [31]) and related versions of this scale (e.g., mechanical VAS, digital VAS). Other commonly used measures of pain intensity [32] are the 0-10 Numerical Rating Scale (NRS), the Verbal Rating Scale (VRS), and the Faces Pain Scale-Revised (FPS-R [33]). Each of these measures is adequately valid and reliable as a measure of pain intensity, and patients in Western countries tend to prefer VRS and NRS over VAS.[32] Pain tolerance is commonly operationalized as the length (in seconds) that an individual is willing to experience an aversive stimulation.[34–36]

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Finally, perceived stress is usually assessed with the: (a) Perceived Stress Scale (PSS [37]), a 10-item scale that asks about people's feelings and thoughts regarding the passing month, on a scale from 0 (never) to 4 (very often); (b) stress scale of Depression, Anxiety, and Stress Scale (DASS [38]), a 14-item scale where subjects are asked to use 4-point severity/frequency scales to rate the extent to which they have experienced each state over the past week; and (c) 21-item version (DASS-21), which consist of a 7-item scale similar to the previous. In contrast, objective stress is usually measured through salivary cortisol, heart rate variability, or blood pressure measurement and analysis.[39–41]

Study selection, data management and data extraction

First, all references identified during the database search will be inserted into Zotero Software.[42] Second, cross-references and duplicates will be deleted. Then, two independent reviewers (MJar and IG) will screen the title, abstract, and keywords of the identified studies for eligibility screening based on the eligibility criteria listed above. Full texts of the manuscripts meeting the eligibility criteria and those with an uncertainty of inclusion from the abstracts will be independently read by the same reviewers to confirm inclusion or exclusion. Discrepancies found will be settled during a consensus meeting. In cases in which consensus is not achieved, a third reviewer will be consulted (AFV). The studies' inclusion or exclusion will be recorded on a self-designed data extraction form in Microsoft Excel 10.

Third, detailed data from the studies meeting the inclusion criteria will be extracted on a Microsoft Excel extraction sheet by two authors (MJar and IG). Data to be extracted include are: (1) authors and year of publication; (2) country of study; (3) study participant (type of pain – chronic or acute – and pain etiology); (4) sample size; (5) study participants age (mean, standard deviation); (6) percentage of female

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participants; (7) site of pain; (8) study participants' religious denomination and self-reported spirituality (e.g., self-proclaimed as "spiritual"); (9) study participants' type of religious practice (religious and a practitioner, religious but not a practitioner); (10) study participants' attitudes toward the religion (negative attitude towards deity vs. positive attitude towards deity); (11) length and intensity (frequency and length of sessions) of prayer-based intervention; (12) type of prayer (*private* or *communal* [i.e., dyadic or group] *participatory* prayer); (13) content of prayer (petition, thanksgiving, adoration, confession, reception, meditative, or ritualistic); (14) level of engagement and emotional involvement with the prayer-based intervention; (15) type and characteristics of the comparator or control condition; (16) self-reported pain intensity; (17) pain tolerance; and (18) pain-related stress. Data (mean, standard deviation, or effect sizes) on the primary outcomes will be extracted. Whenever possible, the rates of meaningful reductions in pain intensity – i.e., from responder analyses – will also be extracted; more specifically, the rates by which different groups obtained a 30% and a 50% reduction in pain intensity. Any discrepancies in the data collected will be resolved by discussion and consensus. In the event that consensus is not achieved, a third author (AFV) will be consulted. Full-text screening and data extraction are planned to be completed in July 2021.**Risk of bias assessment**

The risk of bias of the studies included in the present systematic review will be evaluated based on the CONSORT (Consolidated Standards of Reporting Trials) statement. Risk of bias will be assessed by two independent reviewers (MJar and IG) and discrepancies will be settled during a consensus meeting. If a consensus is not reached, a third reviewer (AFV) will be consulted. Risk of bias assessment is anticipated to be completed in August 2021.

Data analysis plan

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Qualitative synthesis. A qualitative synthesis of the body of research will be conducted,[43] describing the methodological characteristics of the studies included in this systematic review, their strengths and limitations, and each study's results regarding the effects of *private* or *communal participatory* prayer in reducing pain intensity and pain-related stress and in increasing pain tolerance in adults experiencing pain, compared to treatment as usual or no treatment. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be summarized and examined separately. To summarize the evidence, we will implement a narrative summary synthesis method based on the framework developed by the UK Economic and Social Research Council to the conduct of narrative synthesis in systematic reviews,[44] modified to fit our purposes. Following the referred guidance, the synthesis process will involve three elements: (1) developing a preliminary synthesis of findings of included studies regarding the two topics; (2) exploring relationships in the data; and (3) assessing the robustness of the synthesis.

Meta-analysis. Outcomes from *private participatory* prayer and from *communal participatory* prayer will be analyzed separately. If at least two included articles present data on a single primary outcome, then meta-analysis will be performed.[45] End-point scores will be computed as expressed mean differences or standardized mean difference (SMD) for each study for primary outcomes, with associated 95% confidence intervals. Data extracted in the articles will be transformed and presented in Cohen's *d* effect size with 95% confidence interval (*CI*) when possible using the Campbell Collaboration online calculator [46]. The Cohen's *d* effect size thresholds we will use: small (0.20), medium (0.50), large (0.80), and very large (1.30).[47] Data analysis is anticipated to be completed in August 2021.

Assessment of reporting biases

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We will use funnel plots to evaluate the potential reporting bias and small-study effects when there are more than 10 studies in the meta-analysis. Quantification of publication bias will be done using Egger's regression test method.[48] Assessment of reporting biases is anticipated to be completed in August 2021.

Confidence in cumulative evidence

Included studies will be critically appraised and synthesized in terms of methodological quality along with the relevance, strength and limitations of the evidence presented. The similarities and differences between the studies (study characteristics, design and execution) will be described and their potential impact on study outcome will be discussed. The strength of the body of evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

DISCUSSION

The present systematic review will provide evidence for prayer effectiveness in reducing pain intensity and pain-related stress and increasing pain tolerance in adults experiencing acute or chronic pain. This allows for a more considerate and insightful choice when advising pain patients with effective coping strategies.

ETHICS AND DISSEMINATION

Ethics approval is not required for this review because it relies on secondary data. Any modification to the systematic review protocol will be submitted to review and approval of PROSPERO registry and described in the final report of the systematic review. The systematic review findings will be published in a peer-reviewed indexed

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journal and presented at a pain international scientific meeting. It is anticipated that the final report will be completed by September 2021.

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AUTHORS' CONTRIBUTIONS

AFV conceived the study idea, and together with MJar, MJ and MD developed the design of the protocol. IQG, FP, RC, JPR and MJ substantially contributed to the review and refinement of the review design and protocol. AFV and IQG planned the data analysis. AFV and MJar wrote the first draft of the manuscript. All authors critically

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reviewed the manuscript and approved the final version. AFV, MJar and IQG will conduct the review procedures. AFV is the principal investigator, obtained funding and is the guarantor of the review.

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COMPETING INTERESTS

None to declare.

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Appendix 1. Search terms, alternative search terms, and search strategy for Medline via Web of Science

	PAIN	PRAYER-BASED INTERVENTIONS	PAIN-RELATED OUTCOMES
MeSH heading	- Pain - Chronic pain - Acute pain	- Religion - Spiritual therapies	- Pain intensity - Pain tolerance - Stress
Title, abstract, and keywords	- Pain	- Religion - Pray* - Religious* - Spiritual therapies - Noetic therapies	- Pain intensity - Pain severity - Pain tolerance - Stress*

To identify articles related to prayer-based interventions

Pain

AND

Religion OR Pray* OR Religious* OR Spiritual therapies OR Noetic therapies

AND

Pain intensity OR Pain severity OR Pain tolerance OR Stress*

Limits: Adults >18 years; Humans; Randomized Control Trial

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